Wasif Faisal

Payment Security Research • Cyber Threat Intelligence • Protocol Security

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PROFILE

Early-career security researcher specializing in **payment security**, cyber threat intelligence, and transaction protocol analysis. Investigates payment-fraud mechanisms, credential abuse, and authentication vulnerabilities in Card-Not-Present (CNP) and EMV 3-D Secure (3DS) architectures, assessing empirical evidence and translating technical findings into defensible risk recommendations.

EDUCATION

BRAC University — Summer 2026

B.Sc. in Computer Science — Dhaka, Bangladesh

Completed all degree requirements in Summer 2026; formal certificate issuance scheduled for 29 October 2026.

B.Sc. Thesis: *CNP-SHIELD: Investigating Card-Not-Present (CNP) Vulnerabilities in Payment Gateways* (Project Lead & Lead Author).

Thesis Supervisors: Md. Akhtaruzzaman Adnan (Senior Lecturer); S. M. Taiabul Haque, PhD (Professor).

PUBLICATIONS & MANUSCRIPTS

Prefix Count Limits Can Increase First-Hit Discovery in Card Reissuance — **2026**

Under review for Financial Cryptography and Data Security (FC) 2027 • `arXiv:2609.31398` — Lead Author

Discovered structural conditions where prefix caps make active cards substantially easier to enumerate in BIN attacks (guessing valid card numbers); demonstrated the effect via empirical simulations over 50,000 synthetic candidates. The findings equip card issuers and risk teams to rigorously audit reissuance policies for discovery risk.

Payment Authentication Security (EMV 3-D Secure 2.0) — **2025–Present**

Manuscript in preparation for submission — Lead Author

Multi-author faculty-supervised research analyzing trust boundaries, request-binding mechanisms, and attacker-observable flaws across commercial 3DS 2.0 deployments and checkout orchestrations.

RESEARCH EXPERIENCE

Independent Payment Security Researcher — Jan 2018–Present

Self-Directed Research in Payment Fraud & Adversarial Threat Intelligence — Remote

- Track publicly observable payment-abuse ecosystems, analyzing card-testing methodologies, BIN attack tools, and stolen-credential tradecraft across underground communities.
- Evaluate transaction and authentication pipelines to identify missing caller authorization, weak session-transaction binding, and response-handling disclosures.
- Translate complex technical vulnerabilities into structured risk reports and control recommendations for payment product, fraud, and security teams.

SELECTED RESEARCH PROJECTS & ARTIFACTS

EMV 3-D Secure Protocol Lab — **2025–Present**

Interactive Protocol Explorer • `cnpshield.github.io/3dslab` — Public Research Resource

Architected an open, vendor-neutral interactive reference for visualizing, comparing, and debugging browser and app 3DS message flows, trust boundaries, and decision points across published specification versions.

Payment Gateway Flow Analysis & Vulnerability Research — **2024–2026**

Undergraduate Thesis Artifacts • Department of CSE, BRAC University — Applied Research

Investigated merchant-visible payment gateway flows, evaluating response normalization, error code leaks, and card verification behavior to design defensive integration mitigations.

ACADEMIC SERVICE

Artifact Evaluation Program Committee Reviewer — **2026**

ACM Conference on Computer and Communications Security (CCS 2026-B) — Artifact Evaluation Track

Evaluated submitted research software artifacts, verification scripts, and experimental datasets for correctness, reproducibility, and alignment with paper claims, observing strict peer-review confidentiality.

SKILLS & TECHNICAL COMPETENCIES

Payment Security & Risk: Coordinated vulnerability disclosure (CERT/CC, EMVCo, payment vendors), cross-organizational security communication, payment-fraud analysis, BIN attacks, EMV 3DS 2.0 protocol analysis, transaction-risk assessment, and evidence-based technical reporting.

Tools & Verification: Python, SQL/SQLite, formal verification (ProVerif, Tamarin), Burp Suite, browser developer tools, Git, Linux, PowerShell, and LaTeX.

Leadership & Activities: Assistant Director of Marketing (BUNSC), Senior Executive for Research & Publication (BUEDF), Senior Executive for HR (BULDF), Organizing Committee Volunteer (17th Convocation), BRAC University.